Deploy

FoodMythBuster

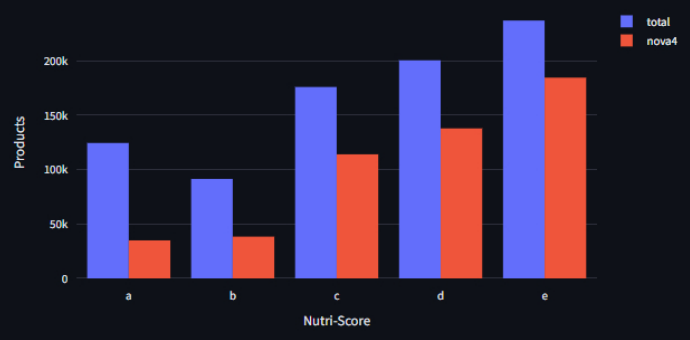
Are **'healthy' food labels** a lie? Investigating whether front-of-pack health claims align with **NOVA classification** (degree of industrial processing).

Total Products	NOVA 4 (Ultra-Processed)	With Health Claim	Deceptive (Claim + NOVA 4)
1,094,802	697,568	194,678	91,955

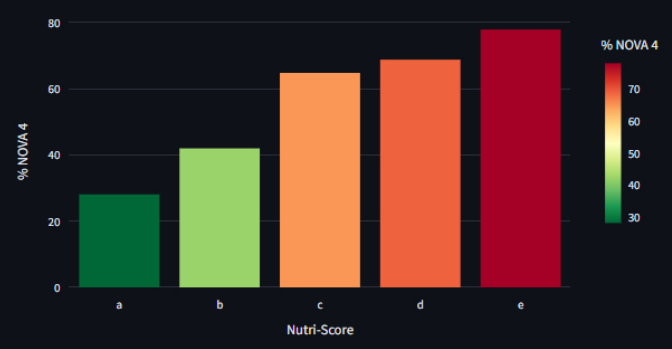
1 · The Paradox: Nutri-Score A/B but Ultra-Processed

What is Nutri-Score? A front-of-pack nutrition label developed in France (2017) and adopted across several European countries. It grades packaged foods from **A (green, best)** to **E (red, worst)** based on nutrient density per 100 g — sugar, salt, saturated fat and calories weighed against fibre, protein, fruits and vegetables. It rates **nutritional composition**, not degree of processing — so a product reformulated to score well on paper can still be **ultra-processed (NOVA 4)**, which is exactly what this chart reveals.

Products by Nutri-Score: Total vs NOVA 4



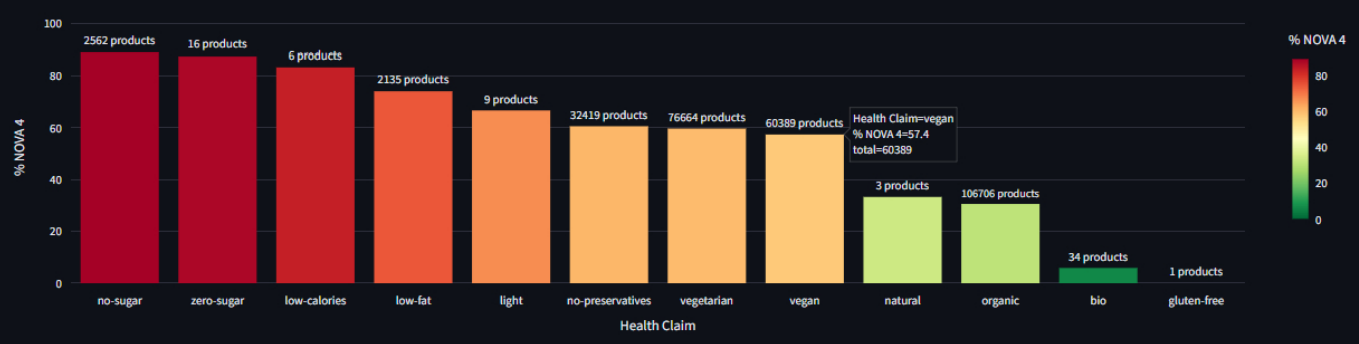
% Ultra-Processed by Nutri-Score Grade



2 · 'Green Label' but Ultra-Processed

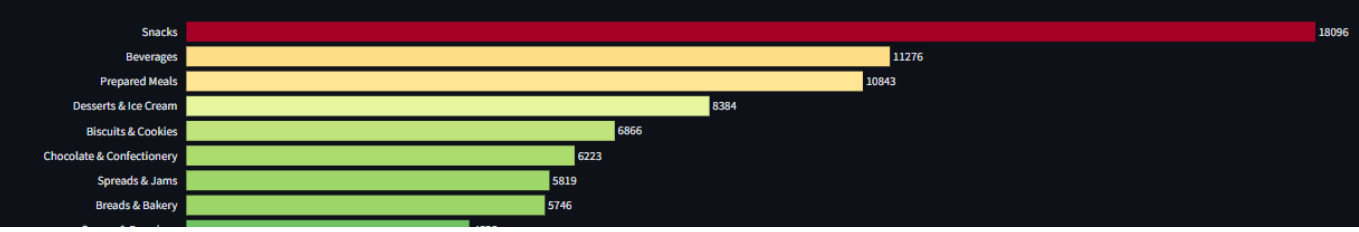
91,955 products in this dataset carry at least one front-of-pack health claim *and* are classified **NOVA 4 (ultra-processed)** — that is **47.2%** of all claim-bearing products. Below, the share of NOVA 4 broken down by each individual claim.

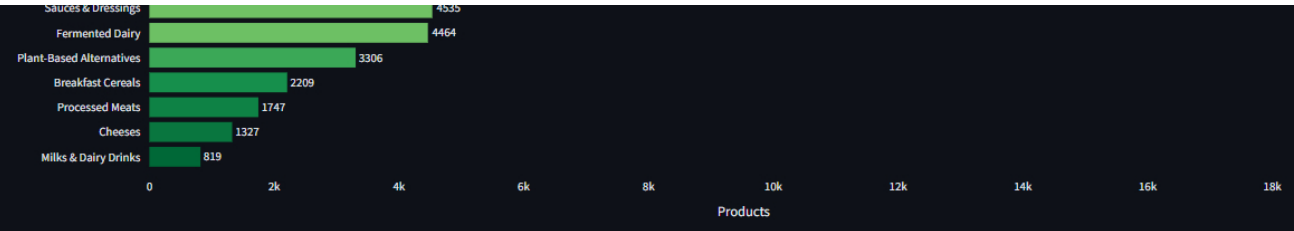
% Ultra-Processed Among Products Carrying Each Health Claim



3 · 'Healthy Junk' by Category Group

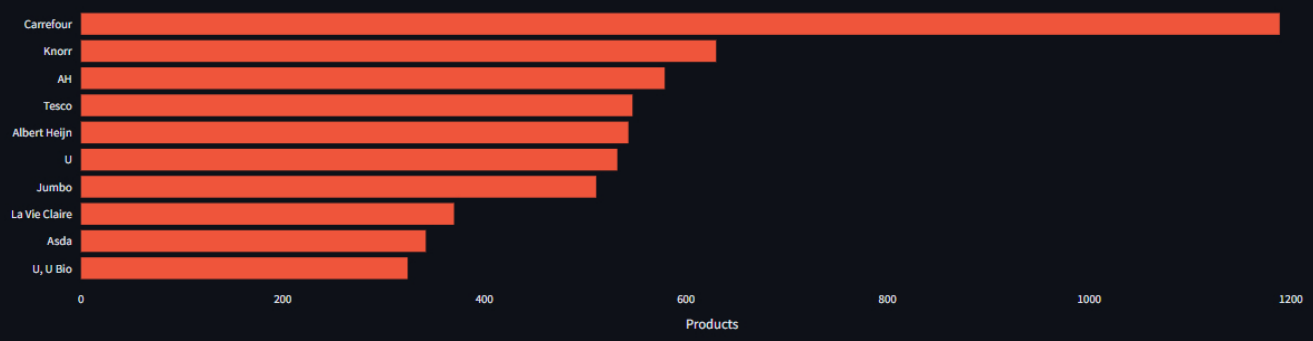
Deceptive Products (Health Claim + NOVA 4) by Category Group





4 · Top Brands: Health Claim + Ultra-Processed

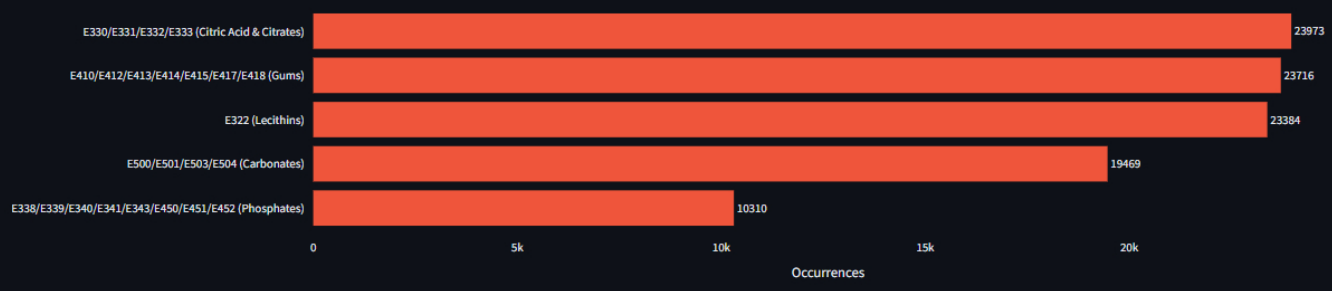
Top 10 Brands Selling NOVA 4 Products Under Health Claims



5 · Top 5 Additive Families in Deceptive Products

E-numbers from products that are **NOVA 4 and carry a health claim**, grouped by functional family. Roman-numeral suffixes (e.g. E450i, E450ii) are collapsed to the base code.

Top 5 Additive Families in NOVA 4 Products with Health Claims



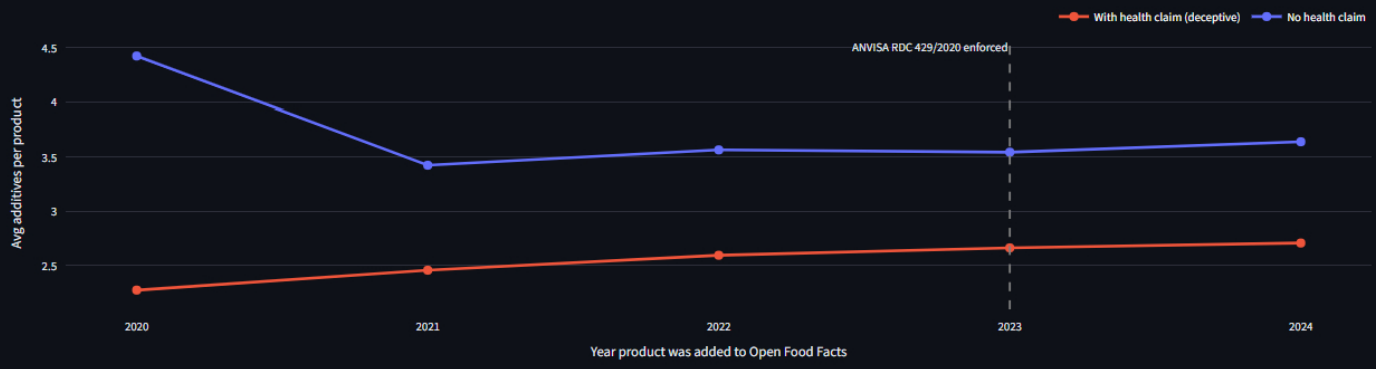
Health concerns linked to these additive families

- **Phosphates** — Concerns for kidney issues and coronary heart disease risk. source

6 · The Widening Gap: Deceptive NOVA 4 Use More Additives Over Time

Among NOVA 4 products, average number of additives per product, split by whether they *also* carry a health claim. The 2023 dashed marker is when **ANVISA RDC 429/2020** mandatory front-of-pack warning labels became enforceable in Brazil — sample sizes jump that year, so read the trend carefully.

Avg additives per product · NOVA 4 with vs without health claim



Sample sizes per year — **2020** → deceptive n=11332, non-deceptive n=125766 · **2021** → deceptive n=9834, non-deceptive n=43701 · **2022** → deceptive n=8280, non-deceptive n=38880 · **2023** → deceptive n=6079, non-deceptive n=28331 ·

2024 → deceptive n=7428, non-deceptive n=39250

Data: Open Food Facts · NOVA classification: Prof. Carlos Monteiro, University of São Paulo · Built for Data Engineering Zoomcamp